Cauchy-Goursat Theorem: If a function f is analytic at all pts interior to and on the curve C (simple closed contour), then

$$\int_C f(z)\,dz = 0$$

Proof:

Lemma: Let f be analytic through a closed region R consisting of pts interior to a positively oriented simple closed curve C together with pts on C. For any $\varepsilon > 0$, R can be covered with a finite number of squares / partial squares with index $j = 1, \dots, n$ s.t. in each one there exists a fixed pt z_j for which $\left| \frac{f(z) - f(z_j)}{z - z_j} - f'(z_j) \right| < \varepsilon$ for all pts in the square (except z_j)

Study 4

$\overline{v_0} > \overline{v_1} > \overline{v_2} > \dots$

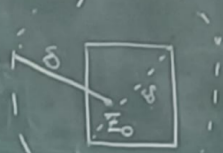
Nested Squares

Then, there exists a point z_0 which is in all of the squares

R

$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} = f'(z_0)$

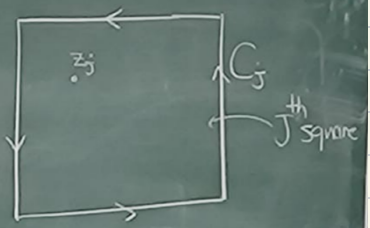
For any $\varepsilon > 0$, $\exists \delta > 0$ s.t. if $|z - z_0| < \delta$, $\left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| < \varepsilon$



There exists $\overline{\square}_k \subset B(z_0, \delta)$

Note that $|\delta_j(z)| < \varepsilon$ $\lim_{z \to z_j} \delta_j(z) = f'(z_j) - f'(z_j) = 0$

So, $\delta_j(z)$ is cont. on the square.



Given $\varepsilon > 0$, consider the covering of R with such squares / p. squares

On the jth square:

$$\delta_j(z) = \begin{cases} \frac{f(z) - f(z_j)}{z - z_j} - f'(z_j) & z \neq z_j \\ 0 & z = z_j \end{cases}$$



$$f(z) = f(z_j) - z_j f'(z_j) + f'(z_j)\cdot z + (z - z_j)\,\delta_j(z)$$

$$\int_{C_j} f(z)\,dz = [f(z_j) - z_j f(z_j)] \underbrace{\int_{C_j} 1\,dz}_{=0} + f'(z_j) \underbrace{\int_{C_j} z\,dz}_{=0} + \int_{C_j} (z - z_j)\,\delta_j(z)\,dz$$

$$\int_{C_j} f(z)\,dz = \int_{C_j} (z - z_j)\,\delta_j(z)\,dz$$

$$\sum_{j=1}^{n} \int_{C_j} f(z)\,dz = \int_{C} f(z)\,dz$$



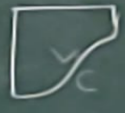
$$\left| \int_{C_j} (z - z_j)\,\delta_j(z)\,dz \right| < \int_{C_j} \left| (z - z_j)\,\delta_j(z) \right| |dz|$$

$|z - z_j| < \sqrt{2}\, s_j$ (s_j = the side length of the square)

$|\delta_j(z)| < \varepsilon$. $\Rightarrow |z - z_j|\,|\delta_j(z)| < \sqrt{2}\cdot s_j\,\varepsilon$.

$$\left| \int_C f(z)\,dz \right| = \left| \sum_{j=1}^{n} \int_{C_j} f(z)\,dz \right| = \left| \sum_{j=1}^{n} \int_{C_j} (z - z_j)\,\delta_j(z)\,dz \right|$$

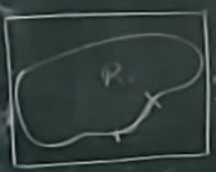
$$\leq \sum_{j=1}^{n} \left| \int_{C_j} (z - z_j)\,\delta_j(z)\,dz \right|$$



$$\left| \int_{C_j} (z - z_j)\, \delta_j(z)\, dz \right| < \sqrt{2}\, s_j \varepsilon\, 4 s_j = 4\sqrt{2}\, A_j \varepsilon$$

s_j = the side of the square

A_j = the area of the square



For a partial square

$$< \sqrt{2}\, s_j \varepsilon (4 s_j + L_j) < 4\sqrt{2} A_j \varepsilon + \sqrt{2} L_j \varepsilon s_j$$

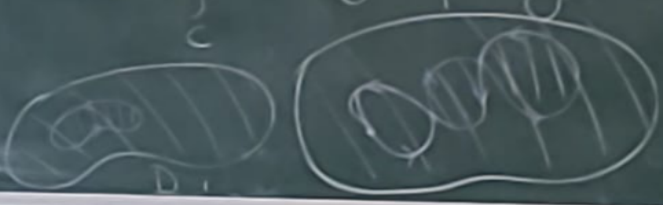
$$(4\sqrt{2} A_j + \sqrt{2} L_j s_j)\varepsilon$$

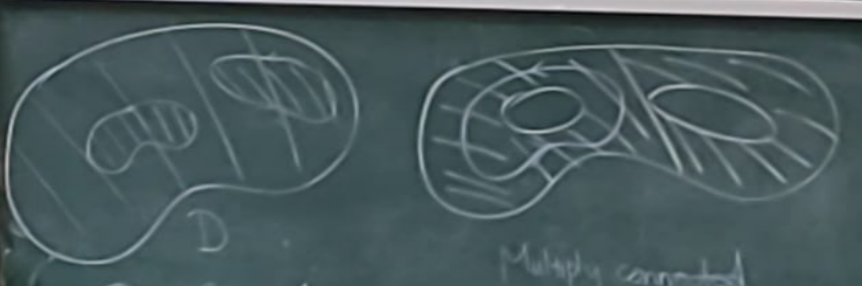
$$\sum_{j=1} (4\sqrt{2} A_j + \sqrt{2} L_j s_j)\varepsilon < (4\sqrt{2} \cdot S + \sqrt{2} \cdot L \cdot s)\varepsilon$$

Simply Connected / Multiply Connected Domains

A simply connected domain D is a domain s.t. every simple closed contour within it encloses only pts in D (Otherwise, it's multiply connected)

Thm: If $f(z)$ is analytic throughout a simply connected domain D, then $\int_C f(z)\, dz = 0$ for every closed curve C in D.

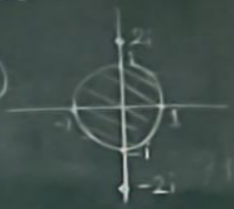




[theorem]

Corollary: If an analytic function on a simply connected domain has an antiderivative

Ex: $\int_C \frac{z^3}{z^2+4}\,dz = 0$



C = the unit circle around (0,0)

Theorem: Suppose that

i) C is a simple closed contour with positive orientation

ii) C_k $k=1,\dots,n$ are simple close contours interior to C with clockwise orientation

If f is analytic on all these curves and on the domain (multiply) consist C outside C_k's Then,

$$\int_C f(z)\,dz + \sum_{k=1}^{n} \int_{C_k} f(z)\,dz = 0$$



$$\int_{\Gamma_1} f(z)\,dz + \int_{\Gamma_2} f(z)\,dz = 0$$